香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

工學院
SCHOOL OF ENGINEERING

Industrial Engineering & Decision Analytics
工業工程及決策分析

IEDA 1250 Lecture 2: Linear Programming

Instructor: Bo YANG

Summer, 2026

Today's Plan

- Motivation
- Solve LPs with two variables
- A brief introduction to the simplex and interior point methods
- Duality theory
- Solve LPs with Excel

Motivation

- Linear programming (LP) is fundamental to modern optimization:
  - Important in its own
  - Commercial solvers (e.g., Gurobi, CPLEX, etc.) can easily handle large scale LPs
  - Many problems can be reduced to LPs
  - Widely used in IE/OR, OM, ECE, CS …

Motivation

- Planning your week: You have **24 * 7 = 168** hours in a week, which is the "resource" you want to allocate wisely
  - Suppose you spend time on the following activities:
    - Study (**S1**)
    - Sleep (**S2**)
    - Social (**S3**)
    - Everything else (**E**)
  - Meanwhile, you need to satisfy the following constraints:
    - **S1 >= 60** (to finish HWs and pass exams)
    - **S2 >= 56** (8 hours sleep for each day)
    - **S3 + S2 + E >= S1** (Study hard; play harder! Too much study requires social activities and sleep)
  - Is it possible to satisfy all constraints? Is there a feasible solution?
    - Yes! **S1 = 60, S2 = 56, S3 = 32, E=20**
- Suppose "happiness" is S1 + 2 * S2 + 3 * S3 + 3 * E. How to maximize your happiness?

Motivation

- Decision variables:
  - Study **(S1)**
  - Sleep **(S2)**
  - Social **(S3)**
  - Everything else **(E)**
- Constraints:
  - **S1 >= 60** (to finish HWs and pass exams)
  - **S2 >= 56** (8 hours sleep for each day)
  - **S3 + S2 + E >= S1** (Study hard; play harder! Too much study requires social activities and sleep)
- Objective function: **Max S1** + 2 * **S2** + 3 * **S3** + 3 * **E**

Motivation

- A linear program formulation:

$$
\begin{aligned}
\max_{S_1,S_2,S_3,E} \quad & S_1 + 2S_2 + 3S_3 + 3E \\
\text{subject to} \quad & S_1 \geq 60 \\
& S_2 \geq 56 \\
& S_3 + S_2 + E \geq S_1 \\
& S_1 + S_2 + S_3 + E = 168 \\
& S_1 \geq 0 \\
& S_2 \geq 0 \\
& S_3 \geq 0 \\
& E \geq 0
\end{aligned}
$$

- Be careful when building an LP model. You need to consider all possible constraints
- Optimal solution: **S1 = 60, S2 = 56, S3 = 26, E = 26, Optimal OBJ = 328**

Linear Programming

- A **linear function** in *n* variables is of the form

$$f(x_1, x_2, \ldots, x_n) = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n$$

where $c_1,\ c_2, \ldots,\ c_n$ are constants

- A **linear inequality** is in one of the forms

$$g(x_1, x_2, \ldots, x_n) \le b \quad \text{or} \quad g(x_1, x_2, \ldots, x_n) \ge b$$

where $g(\cdot)$ is a linear function and b is a constant

- A **linear program** maximizes/minimizes a linear function subject to a set of constraints given by linear inequalities/equalities

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Example: The Galaxy Ltd.

- Galaxy manufactures two toy doll models:
  - Space Ray ($8 profit per dozen) and Zapper ($5 profit per dozen)
- Resources available for each week:
  - 1000 pounds of special plastic
  - 40 hours of production time
- Marketing requirements:
  - Total production cannot exceed 700 dozens per week
  - The production quantity of Space Rays cannot exceed that of Zappers by more than 350 per week
- Technological inputs:
  - Space Rays requires 2 pounds of plastic and 3 minutes of labor per dozen
  - Zappers requires 1 pound of plastic and 4 minutes of labor per dozen

Example: The Galaxy Ltd.

- The current production plan is to:
  - Produce as many as possible of the more profitable product, i.e., Space Ray ($8 profit per dozen)
  - Use leftover resources to produce Zappers ($5 profit per dozen)

- The current production plan consists of:
  - Space Rays = 450 dozens
  - Zapper = 100 dozen
  - Total profit = $4100 per week

- The current production plan is feasible, but is it optimal?

Example: The Galaxy Ltd.

- Decisions variables:
  - x_1 = Weekly production level of Space Rays (in dozens)
  - x_2 = Weekly production level of Zappers (in dozens)
- Objective Function (maximization):
  - Weekly profit: $8x_1+5x_2$
- LP Formulation:

$$
\begin{array}{lll}
\text{Max} & 8x_1 + 5x_2 & \text{(Weekly profit)} \\
\text{subject to} & & \\
 & 2x_1 + x_2 \le 1000 & \text{(Plastic)} \\
 & 3x_1 + 4x_2 \le 2400 & \text{(Production Time)} \\
 & x_1 + x_2 \le 700 & \text{(Total production)} \\
 & x_1 - x_2 \le 350 & \text{(Mix)} \\
 & x_i \ge 0,\ i = 1,2 & \text{(Nonnegativity)}
\end{array}
$$

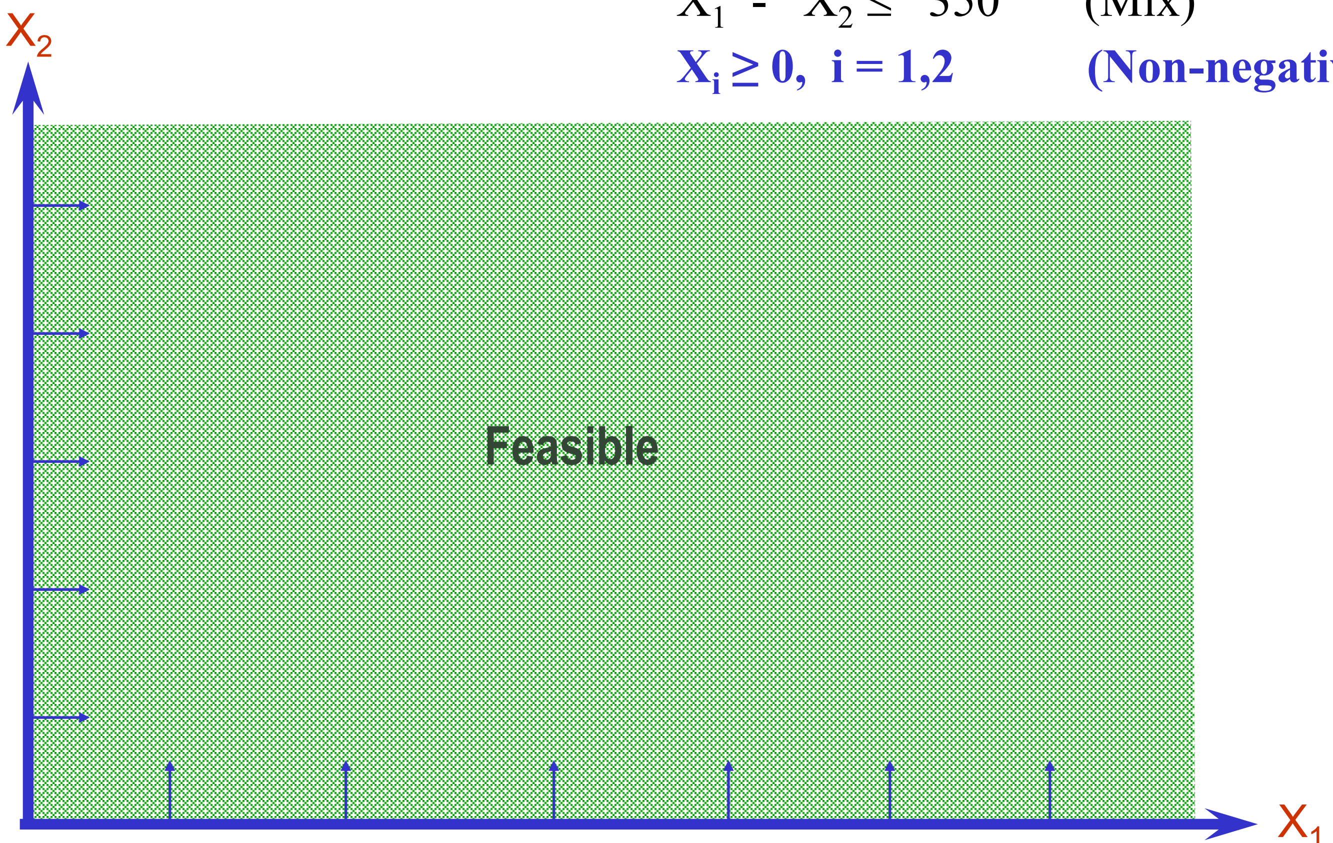
Example: The Galaxy Ltd.

- The set of all points that satisfy all the constraints of the models is called a **feasible region**

- Using the graphical approach, we can represent all the constraints, the objective function, and identify three types of feasible points

Example: The Galaxy Ltd.

Max	$8X_1 + 5X_2$	(Weekly profit)
subject to	$2X_1 + X_2 \le 1000$	(Plastic)
	$3X_1 + 4X_2 \le 2400$	(Production Time)
	$X_1 + X_2 \le 700$	(Total production)
	$X_1 - X_2 \le 350$	(Mix)
	$\mathbf{X_i \ge 0,\ i = 1,2}$	**(Non-negativity)**



Example: The Galaxy Ltd.

Max	$8X_1 + 5X_2$	(Weekly profit)
subject to	$\mathbf{2X_1 + X_2 \leq 1000}$	**(Plastic)**
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	$X_1 + X_2 \leq 700$	(Total production)
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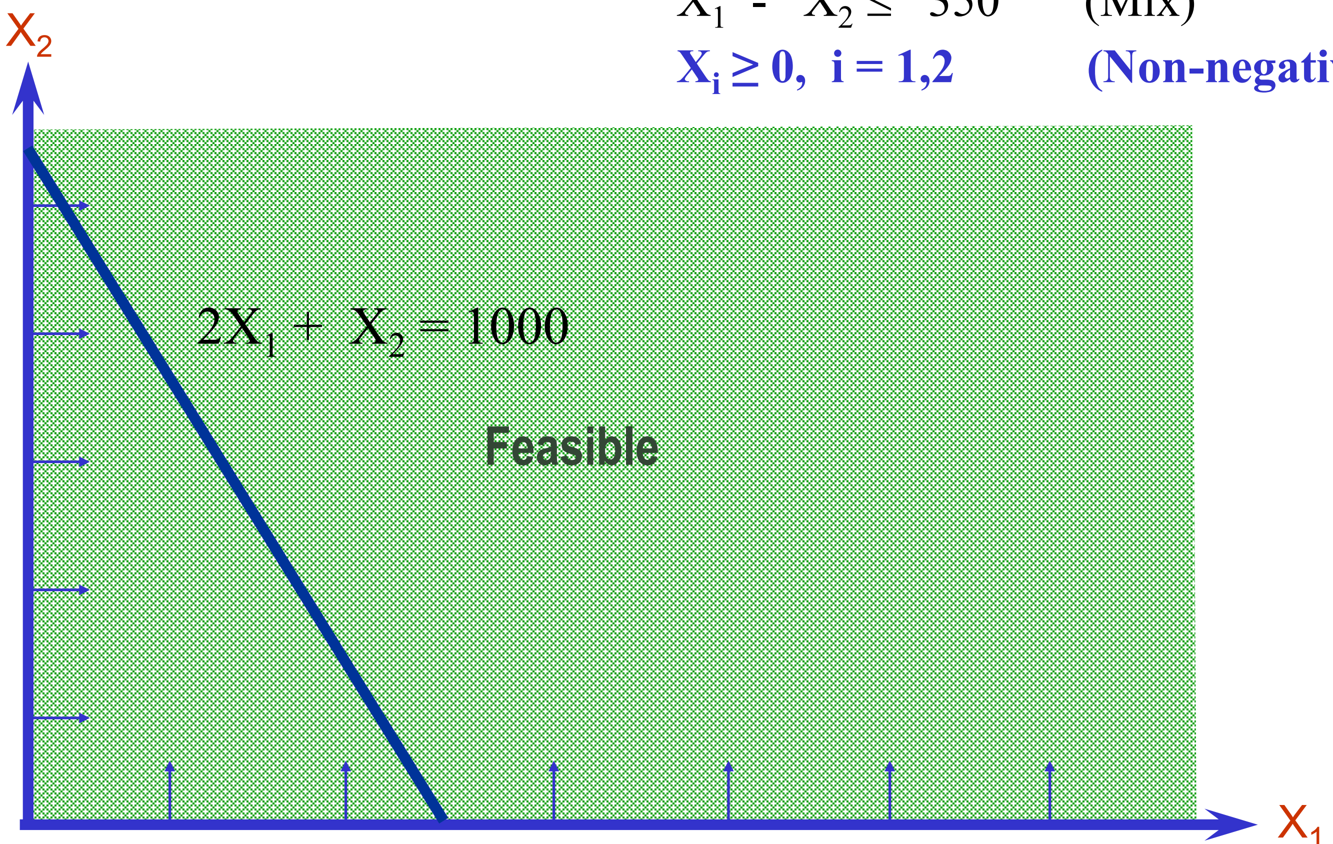

Example: The Galaxy Ltd.

Max	$8X_1 + 5X_2$	(Weekly profit)
subject to	**$2X_1 + X_2 \le 1000$**	**(Plastic)**
	$3X_1 + 4X_2 \le 2400$	(Production Time)
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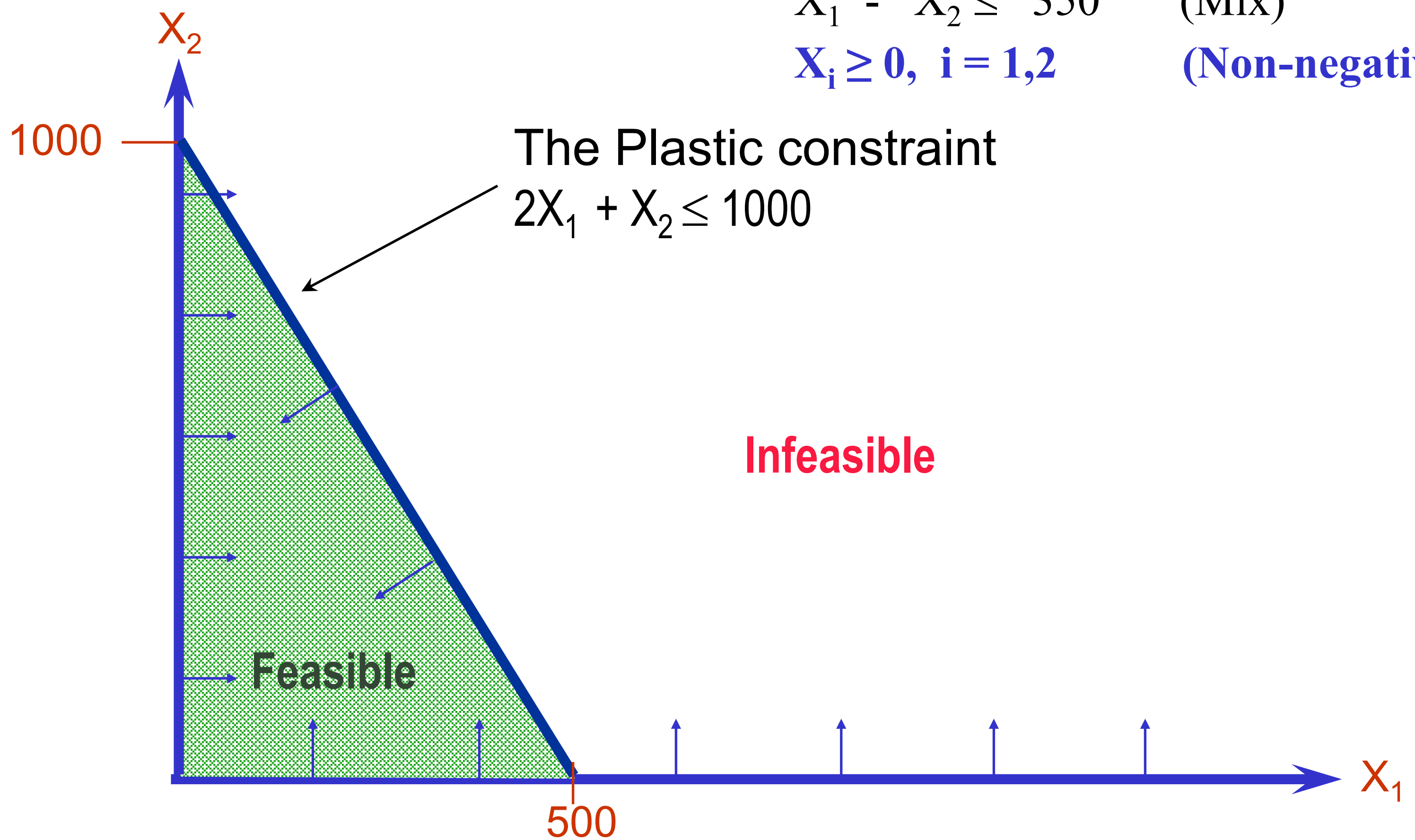

Example: The Galaxy Ltd.

Max	$8X_1 + 5X_2$	(Weekly profit)
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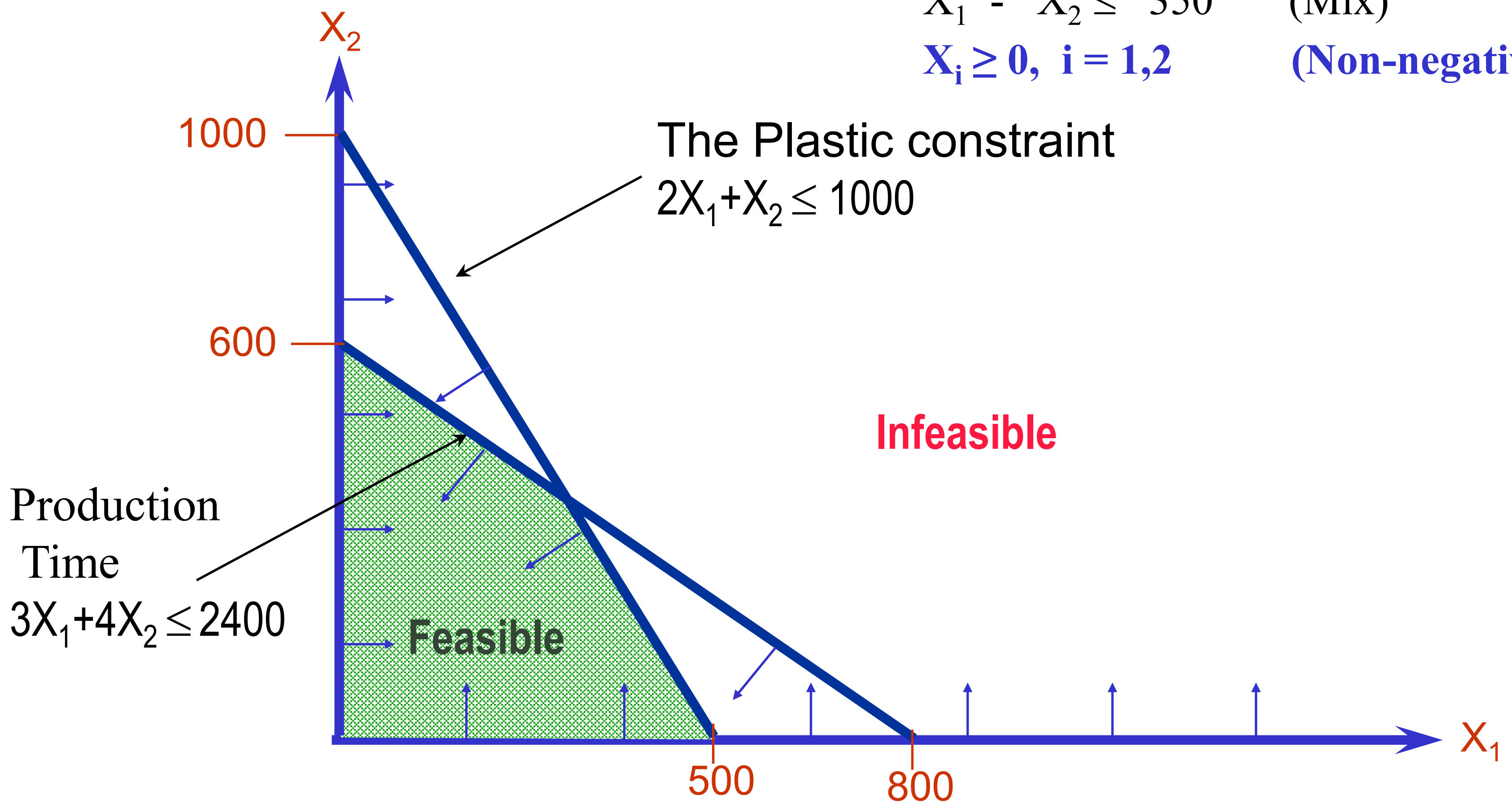

Example: The Galaxy Ltd.

Max	$8X_1 + 5X_2$	(Weekly profit)
subject to	$\mathbf{2X_1 + X_2 \le 1000}$	**(Plastic)**
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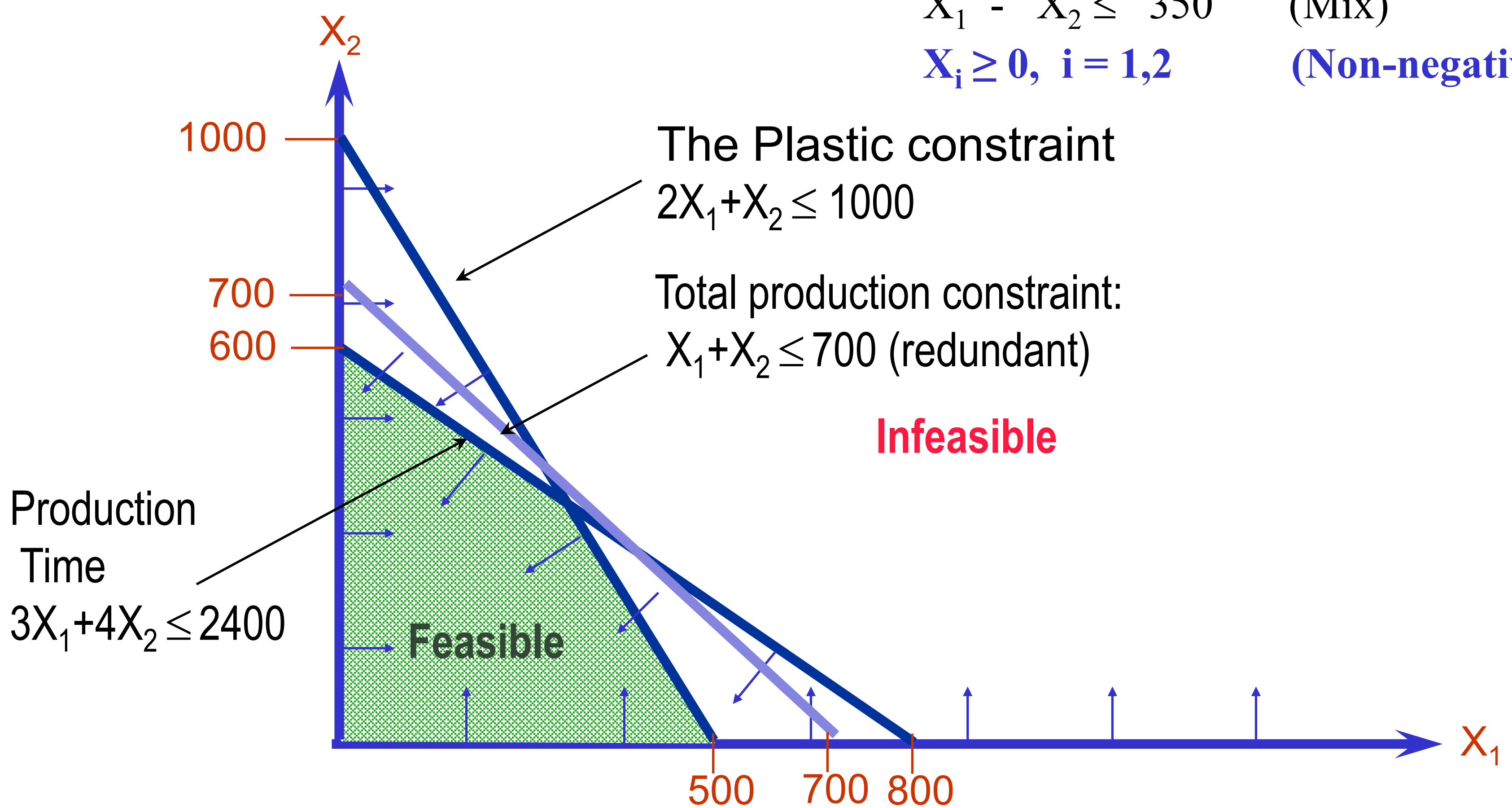

Example: The Galaxy Ltd.

Max	$8X_1 + 5X_2$	(Weekly profit)
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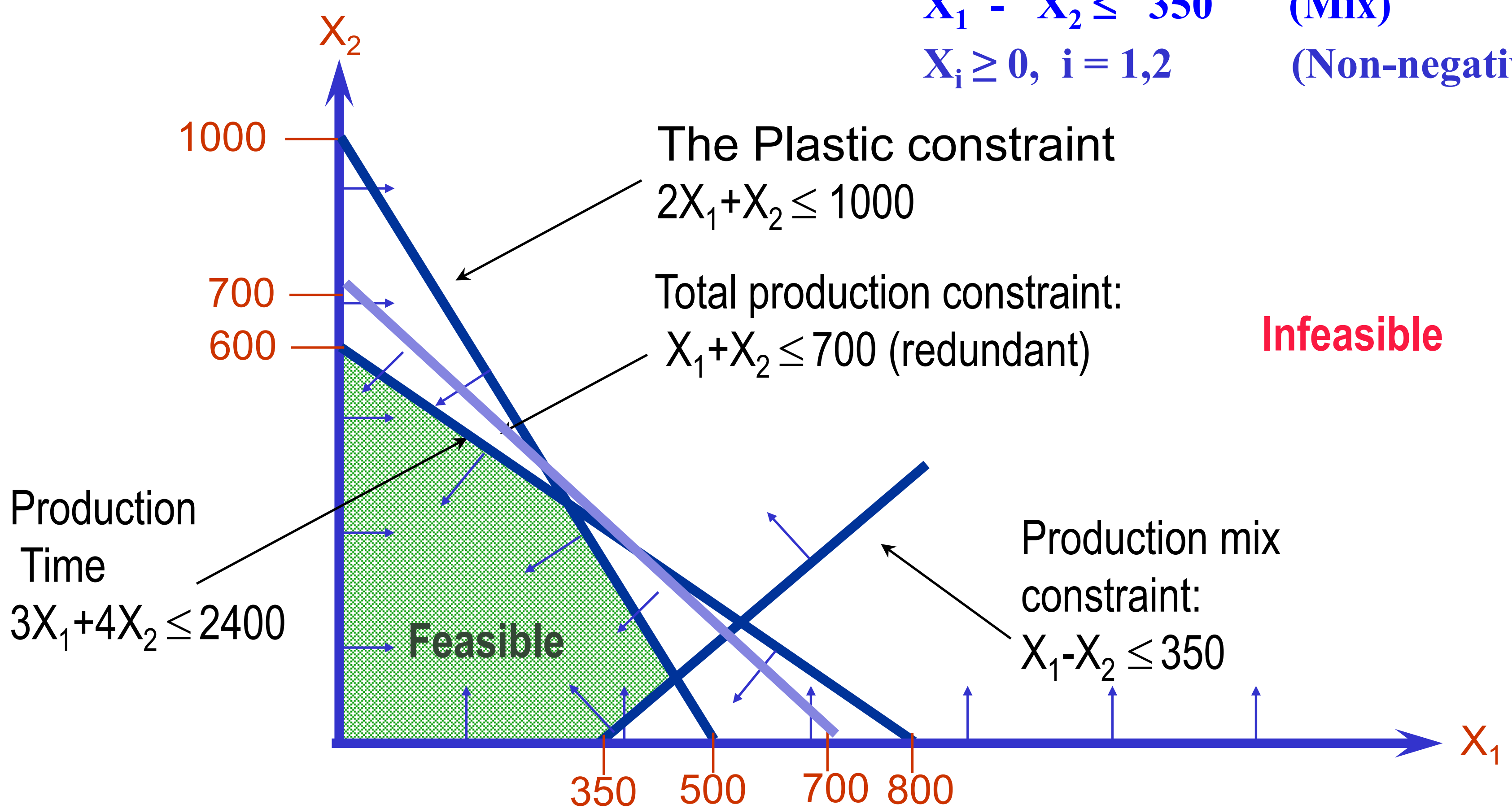
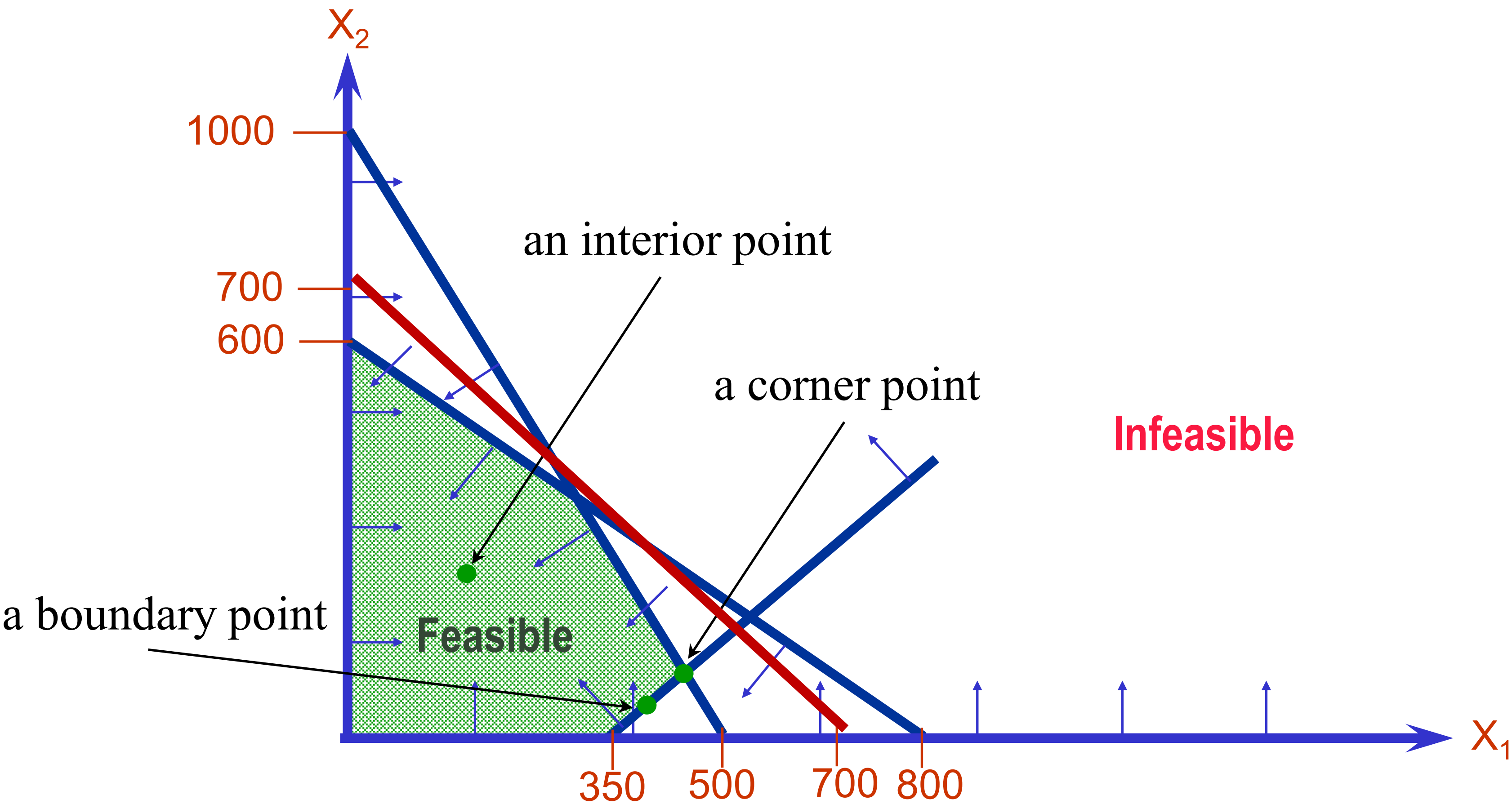


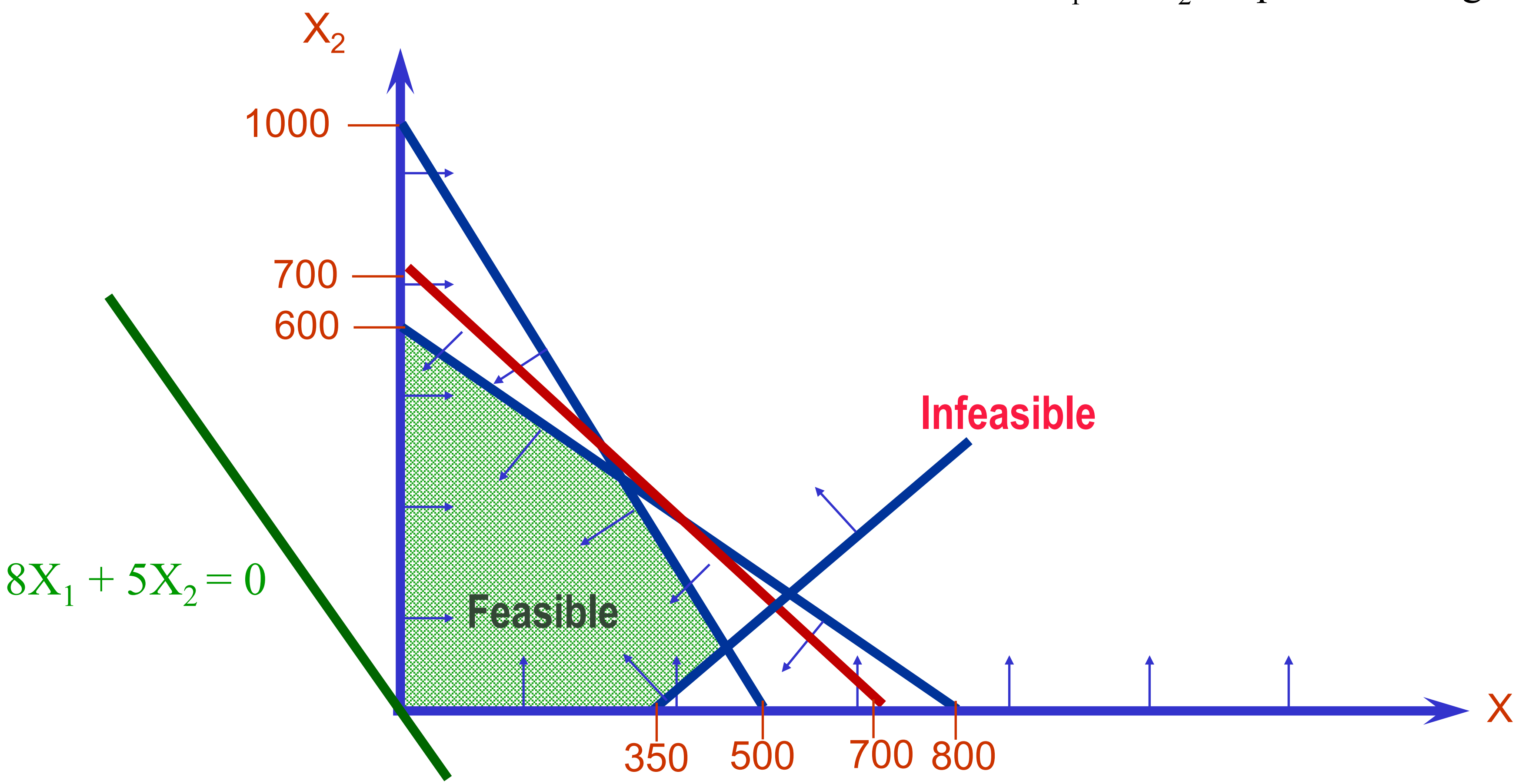
Example: The Galaxy Ltd.

- A feasible point is in the interior or on the boundary or at a corner
- There may be some redundant constraints (e.g., the red line segment)



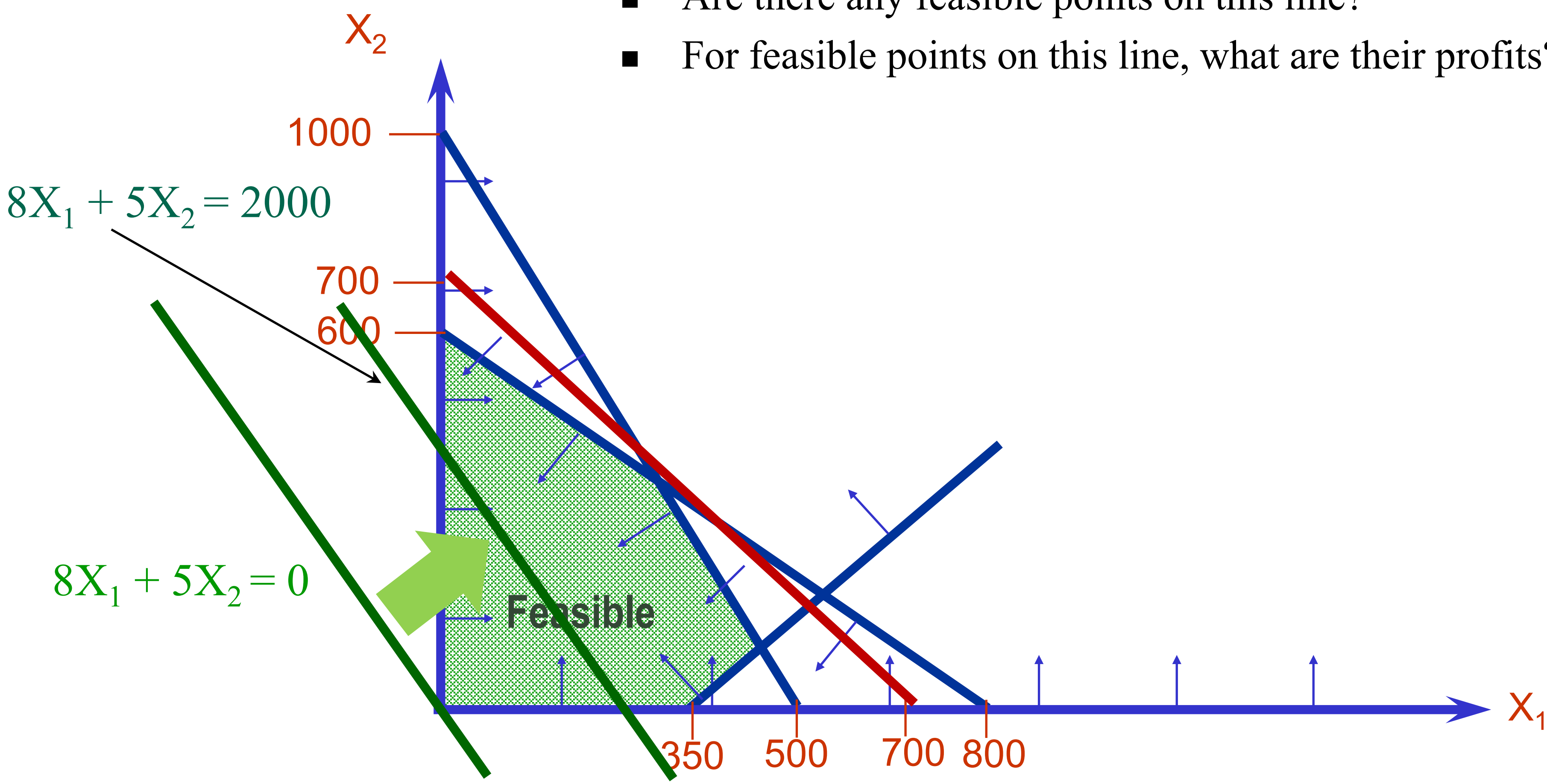
Example: The Galaxy Ltd.

- We still need to maximize our objective function **$8X_1 + 5X_2$**
- Note that the line $8X_1 + 5X_2 = 0$ passes through (0,0)



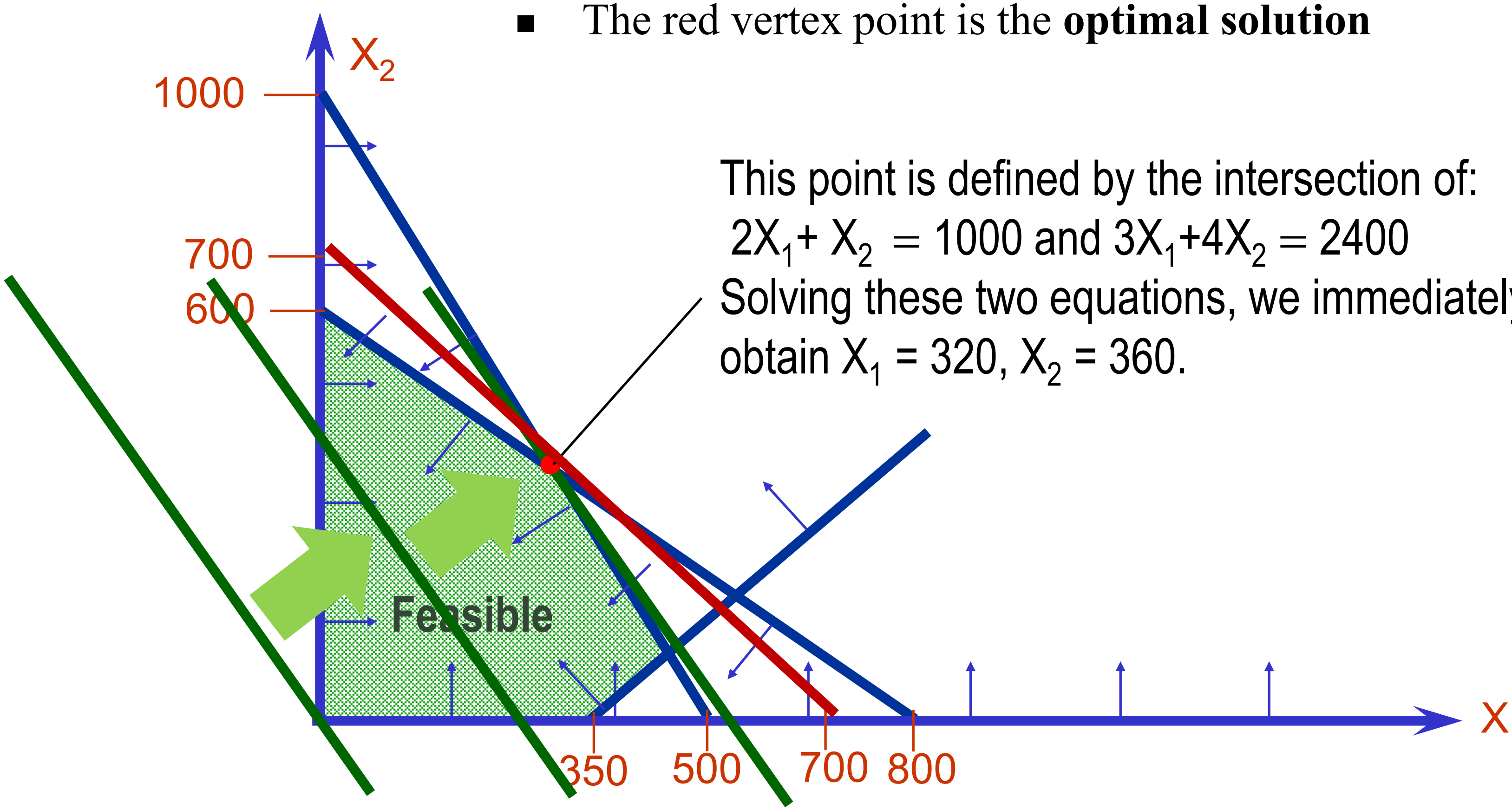
Example: The Galaxy Ltd.

- If we move the line $8X_1 + 5X_2$ to the upper right corner, how does the profit change? Why?
- Are there any feasible points on this line?
- For feasible points on this line, what are their profits?



Example: The Galaxy Ltd.

- Can we further increase the profit beyond $2000?
- How far can we go?
- Why shall we stop at the vertex point highlighted in red?
- The red vertex point is the **optimal solution**



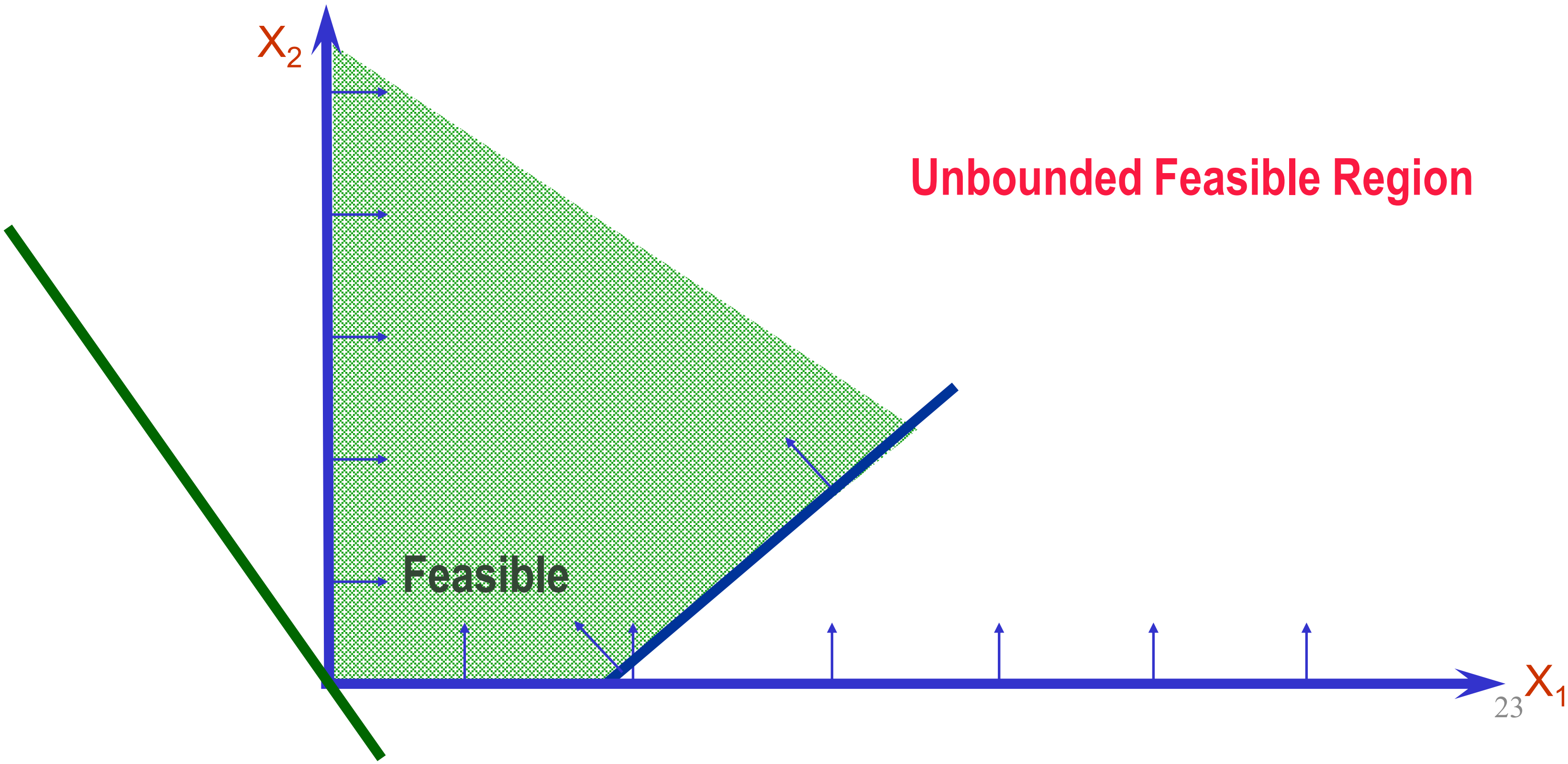
This point is defined by the intersection of:
$2X_1 + X_2 = 1000$ and $3X_1 + 4X_2 = 2400$
Solving these two equations, we immediately obtain $X_1 = 320$, $X_2 = 360$.

Example: The Galaxy Ltd.

- Optimal solution
  - x_1 = 320 dozen
  - x_2 = 360 dozen
  - Profit = 8 * 320 + 5 * 360 = $4360 (> $4100)
- The "plastic" and the production time constraints are tight
  - Thes optimal solution utilizes all plastic and production hours.
- Total production is only 680 (not 700).
- Space Rays production exceeds Zappers production by only 40 dozens.

Revisit the example

- For a linear program formulation, do we always end up with the optimal solution?
  - Does the optimal solution always exist?
  - No, the LP can be unbounded or infeasible



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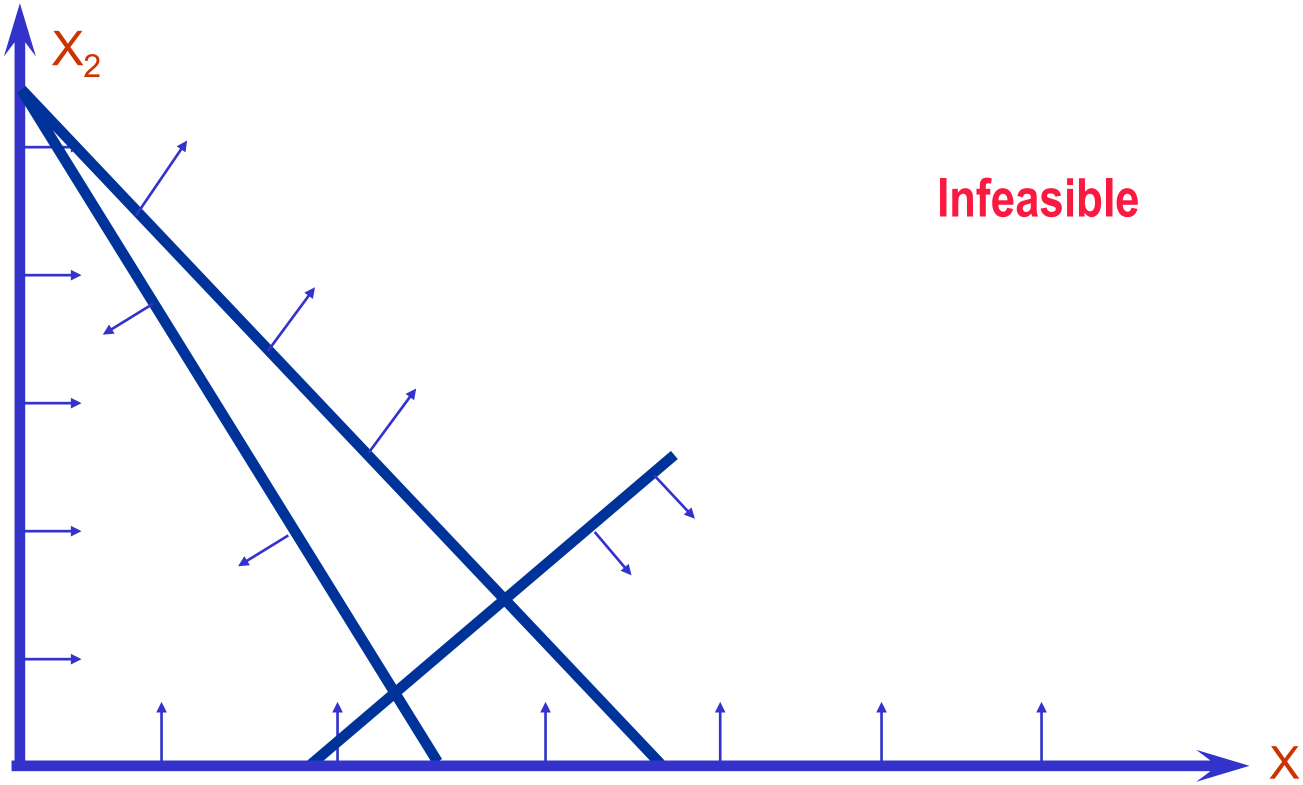

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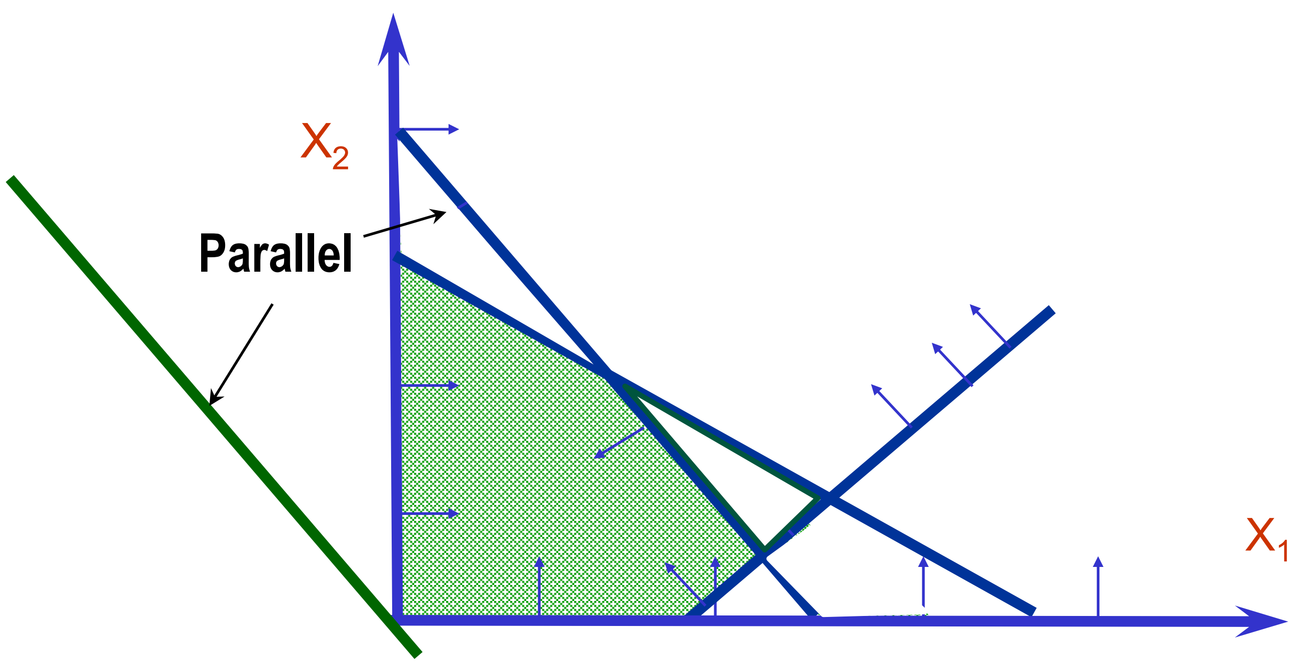
  - No, there may be multiple (or even infinite) optimal solutions



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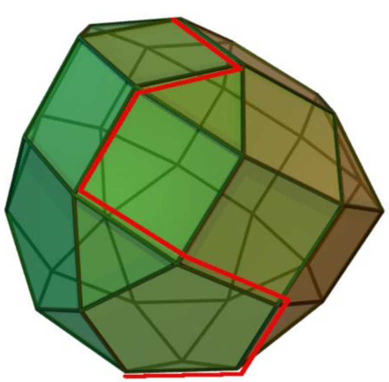
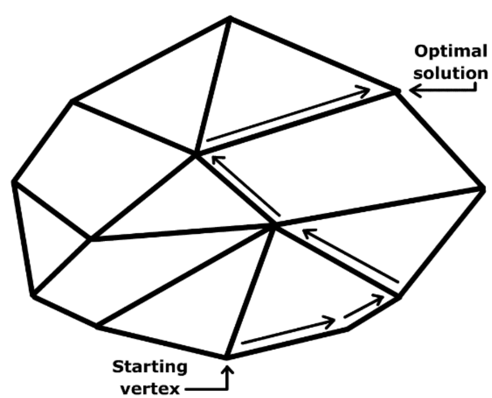
- **For an LP model, the optimal solution is always at a vertex or on the boundary**

Simplex and interior point methods

- The graphical approach is very limited:
  - It can only handle problems with two variables
  - What if we have three dimensional or even N dimensional problems?
  - What if we have hundreds or even millions of constraints?
- Simplex method (Dantzig 1940's)
  - Uses the geometry of the feasible region (polyhedron)
  - Starts from an arbitrary corner point (vertex) and moves to an adjacent corner point (vertex) if such point improves the objective function value most
  - Always returns an optimal solution because there are finite corner points





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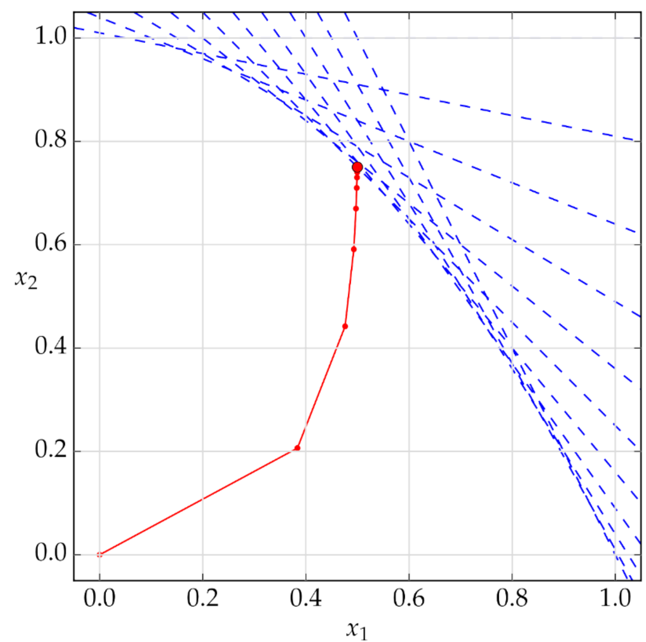
  - Uses the geometry of the feasible region (polyhedron)
  - Starts from an arbitrary corner point (vertex) and moves to an adjacent corner point (vertex) if such point improves the objective function value most
  - Always returns an optimal solution because there are finite corner points
- Problem: there may be exponential number of vertices
  - As the numbers of variables and constraints increase, the number of vertices becomes huge
  - Example: With 50 variables and 100 constraints, there can be roughly 10^{50} vertices
  - Slow in theory, fast in practice, particularly for problems with small to medium sizes
  - The memory requirement is low

Simplex and interior point methods

- Interior point methods
  - Ellipsoid method (Khachiyan 1980): Fast in theory, horrible in practice!
  - Primal dual interior point method: Fast in both theory and practice!
  - It can be used to solve other convex optimization problems
  - Memory requirement is high; There may be numerical issues (ill conditioning)



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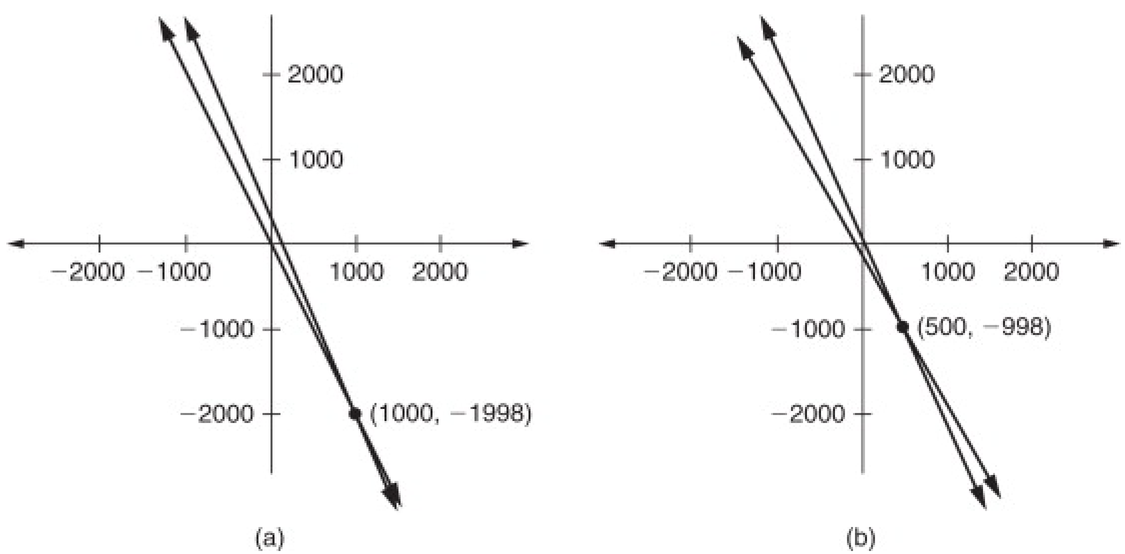
$$(A)\begin{cases} 2x_1 + x_2 = 2 \\ 2.005x_1 + x_2 = 7 \end{cases} \quad \text{and} \quad (B)\begin{cases} 2x_1 + x_2 = 2 \\ 2.01x_1 + x_2 = 7 \end{cases}.$$



Commercial Solvers

- Modern commercial solvers are powerful
  - Gurobi
  - CPLEX
  - COPT
  - AMPL
  - Excel
  - …

- Most of them require programming/coding skills
  - Python
  - Java
  - C/C++
  - …

Commercial Solvers: Gurobi

C:\Users\yangb\source\repos'

```
Academic license - for non-commercial use only - expires 2025-10-26
Set parameter Method to value 2
Set parameter Crossover to value 0
Adding Constraints
Gurobi Optimizer version 11.0.3 build v11.0.3rc0 (win64 - Windows 11+.0 (26100.2))

CPU model: Snapdragon(R) X 12-core X1E80100 @ 3.40 GHz, instruction set [SSE2]
Thread count: 12 physical cores, 12 logical processors, using up to 12 threads

Optimize a model with 115000 rows, 60110 columns and 720000 nonzeros
Model fingerprint: 0x3abbd72f
Coefficient statistics:
  Matrix range     [2e-06, 2e+00]
  Objective range  [2e-04, 2e-04]
  Bounds range     [0e+00, 0e+00]
  RHS range        [1e+00, 3e+01]
Presolve removed 60000 rows and 5000 columns
Presolve time: 0.75s
Presolved: 55000 rows, 55110 columns, 655000 nonzeros
Ordering time: 0.18s

Barrier statistics:
 Dense cols : 110
 Free vars  : 110
 AA' NZ     : 6.000e+05
 Factor NZ  : 4.999e+06 (roughly 90 MB of memory)
 Factor Ops : 4.914e+08 (less than 1 second per iteration)
 Threads    : 12
```

Commercial Solvers: Gurobi

```
                  Objective                Residual
Iter       Primal          Dual         Primal    Dual     Compl     Time
   0   7.82240838e+02  0.00000000e+00  2.04e+03 0.00e+00  4.08e+02     1s
   1   4.98433785e+02  1.03299466e+01  5.95e+02 4.88e-02  1.83e+02     2s
   2   4.30040146e+02  1.24814734e+01  1.39e+02 1.87e-02  7.48e+01     2s
   3   3.64370848e+02  8.81738282e+00  2.55e+01 8.64e-03  3.50e+01     2s
   4   2.89432931e+02  5.20132099e+00  5.55e+00 2.54e-03  1.12e+01     2s
   5   1.79239661e+02  4.24952715e+00  1.92e+00 1.23e-03  5.24e+00     2s
   6   1.00066950e+02  3.85848304e+00  8.16e-01 7.11e-04  2.77e+00     3s
   7   5.07245653e+01  3.50523463e+00  2.40e-01 1.44e-04  7.50e-01     3s
   8   3.43665641e+01  3.57305344e+00  1.33e-01 1.04e-04  4.78e-01     3s
   9   2.52505966e+01  3.87764875e+00  7.97e-02 5.16e-05  2.75e-01     3s
  10   1.93200587e+01  4.44260203e+00  4.97e-02 2.28e-05  1.61e-01     3s
  11   1.68031677e+01  4.98480502e+00  3.82e-02 1.35e-05  1.18e-01     3s
  12   1.46125262e+01  5.12286825e+00  2.85e-02 1.21e-05  9.37e-02     4s
  13   1.29737899e+01  5.59529615e+00  2.02e-02 8.59e-06  7.01e-02     4s
  14   1.22524138e+01  6.16362296e+00  1.62e-02 5.76e-06  5.61e-02     4s
  15   1.14586683e+01  6.47453941e+00  1.24e-02 4.64e-06  4.54e-02     4s
  16   1.09136411e+01  7.07654352e+00  9.65e-03 3.02e-06  3.44e-02     4s
  17   1.05900126e+01  7.42562310e+00  7.66e-03 2.34e-06  2.81e-02     4s
  18   1.04396497e+01  7.75214821e+00  6.64e-03 1.81e-06  2.38e-02     5s
  19   1.02342788e+01  8.12320052e+00  5.31e-03 1.28e-06  1.86e-02     5s
  20   1.01017225e+01  8.41664267e+00  4.21e-03 9.22e-07  1.48e-02     5s
  21   9.98439819e+00  8.63871009e+00  3.29e-03 6.87e-07  1.18e-02     5s
  22   9.91174341e+00  8.77481561e+00  2.68e-03 5.63e-07  9.93e-03     5s
  23   9.86263557e+00  8.91048004e+00  2.19e-03 4.53e-07  8.31e-03     5s
```

```
  33   9.64828642e+00  9.60078931e+00  7.35e-05 1.53e-08  4.13e-04     7s
  34   9.64552629e+00  9.61753142e+00  4.23e-05 8.16e-09  2.43e-04     7s
  35   9.64304873e+00  9.63038936e+00  1.43e-05 3.72e-09  1.10e-04     7s
  36   9.64245710e+00  9.63571006e+00  7.80e-06 1.79e-09  5.86e-05     8s
  37   9.64207975e+00  9.63822298e+00  3.72e-06 9.78e-10  3.35e-05     8s
  38   9.64193339e+00  9.63922880e+00  2.09e-06 6.65e-10  2.35e-05     8s
  39   9.64179920e+00  9.64062983e+00  7.64e-07 2.76e-10  1.02e-05     8s
  40   9.64175386e+00  9.64112738e+00  3.16e-07 1.46e-10  5.44e-06     8s
  41   9.64173258e+00  9.64144504e+00  1.17e-07 5.08e-11  2.49e-06     8s
  42   9.64172137e+00  9.64169444e+00  2.43e-08 6.90e-12  2.35e-07     8s
  43   9.64171920e+00  9.64171725e+00  7.63e-09 4.37e-12  1.60e-08     8s
  44   9.64171820e+00  9.64171848e+00  2.68e-08 1.05e-11  2.51e-10     9s
  45   9.64171820e+00  9.64171816e+00  1.46e-09 2.41e-12  2.51e-13     9s

Barrier solved model in 45 iterations and 8.77 seconds (2.79 work units)
Optimal objective 9.64171820e+00
```

Duality Theory in LP: The Furniture Factory Problem

- A data scientist is in charge of developing the Weekly Production Plan of two key products that the furniture factory makes: chairs and tables. The data scientist predicts that the selling price of a chair is \$45 and the selling price of a table is \$80 dollars.

- There are two critical resources in the production of chairs and tables:
  - Mahogany (measured in board square-feet) and labor (measured in work hours)
  - There are 400 units of mahogany available at the beginning of each week
  - There are 450 units of labor available during each week.

- The data scientist estimates that
  - One chair requires 5 units of mahogany and 10 units of labor.
  - One table requires 20 units of mahogany and 15 units of labor.

- The marketing department has told the data scientist that ALL the production of chairs and tables can be sold.

Duality Theory in LP: The Furniture Factory Problem

- Suppose we can produce and sell fractional quantities of chairs and tables, then the LP formulation is

DATA	CHAIR (x_1)	TABLE (x_2)	CAPACITY
	5 units	20 units	400 units
	10 hours	15 hours	450 hours
PRICE $	$45	$80	

$$
\begin{aligned}
\max_{x_1,x_2} \quad & 45x_1 + 80x_2 \\
s.t. \quad & 5x_1 + 20x_2 \leq 400 \\
& 10x_1 + 15x_2 \leq 450 \\
& x_1, x_2 \geq 0
\end{aligned}
$$

- Suppose we use μ_1 and μ_2 to represent to the opportunity costs of the mahogany and labor, respectively, then the costs of making chairs and tables are at least their prices. The objective is to minimize the cost

		x_1	x_2	
	DATA	CHAIR	TABLE	CAPACITY
μ_1		5 units	20 units	400 units
μ_2		10 hours	15 hours	450 hours
	PRICE $	$45	$80	

$$
\begin{aligned}
\max_{x_1,x_2} \quad & 45x_1 + 80x_2 \\
s.t. \quad & 5x_1 + 20x_2 \le 400 \\
& 10x_1 + 15x_2 \le 450 \\
& x_1, x_2 \ge 0
\end{aligned}
$$

$$
\begin{array}{ccc}
 & & \text{Min} \\
5\mu_1 & 20\mu_1 & 400\mu_1 \\
+ & + & + \\
10\mu_2 & 15\mu_2 & 450\mu_2 \\
\geq & \geq & \\
45 & 80 &
\end{array}
$$

- Suppose we use μ_1 and μ_2 to represent to the opportunity costs of the mahogany and labor, respectively, then the costs of making chairs and tables are at least their prices. The objective is to minimize the cost

	DATA	CHAIR (x_1)	TABLE (x_2)	CAPACITY
μ_1		5 units	20 units	400 units
μ_2		10 hours	15 hours	450 hours
	PRICE $	$45	$80	

$$\begin{aligned} \max_{x_1, x_2} \quad & 45x_1 + 80x_2 \\ s.t. \quad & 5x_1 + 20x_2 \le 400 \\ & 10x_1 + 15x_2 \le 450 \\ & x_1, x_2 \ge 0 \end{aligned}$$

$$\begin{array}{ccc} & & \text{Min} \\ 5\mu_1 & 20\mu_1 & 400\mu_1 \\ + & + & + \\ 10\mu_2 & 15\mu_2 & 450\mu_2 \\ \ge & \ge & \\ 45 & 80 & \end{array}$$

Same optimal objective value!

Duality Theory in LP: The Furniture Factory Problem

- The primal and dual formulations:

$$\begin{aligned}\max_{x_1,x_2} \quad & 45x_1 + 80x_2 \\ s.t. \quad & 5x_1 + 20x_2 \le 400 \\ & 10x_1 + 15x_2 \le 450 \\ & x_1, x_2 \ge 0\end{aligned}$$

$$\begin{aligned}\min_{\mu_1,\mu_2} \quad & 400\mu_1 + 450\mu_2 \\ s.t. \quad & 5\mu_1 + 10\mu_2 \ge 45 \\ & 20\mu_1 + 15\mu_2 \ge 80 \\ & \mu_1, \mu_2 \ge 0\end{aligned}$$

- Properties:
  - (Strong Duality) The primal and dual problems have an identical **optimal objective function value (but different optimal solutions)**
  - (Weak Duality) For any feasible solutions, the dual objective value (minimization) is no smaller than the primal objective value (maximization)
  - The dual of the dual problem is the primal problem
  - **At optimality**, the optimal dual variables are the shadow prices of the resources
  - **At optimality**, complementary slackness (CS) conditions hold

Duality Theory in LP: The Furniture Factory Problem

- The primal and dual formulations:

$$\begin{aligned} \max_{x_1, x_2} \quad & 45x_1 + 80x_2 \\ s.t. \quad & 5x_1 + 20x_2 \le 400 \\ & 10x_1 + 15x_2 \le 450 \\ & x_1, x_2 \ge 0 \end{aligned}$$

$$\begin{aligned} \min_{\mu_1, \mu_2} \quad & 400\mu_1 + 450\mu_2 \\ s.t. \quad & 5\mu_1 + 10\mu_2 \ge 45 \\ & 20\mu_1 + 15\mu_2 \ge 80 \\ & \mu_1, \mu_2 \ge 0 \end{aligned}$$

- Properties:
  - (Strong Duality) The primal and dual problems have an identical **optimal objective function value (but different optimal solutions)**

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$$\begin{aligned} \min_{\mu_1,\mu_2} \quad & 400\mu_1 + 450\mu_2 \\ s.t. \quad & 5\mu_1 + 10\mu_2 \geq 45 \\ & 20\mu_1 + 15\mu_2 \geq 80 \\ & \mu_1, \mu_2 \geq 0 \end{aligned}$$

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Duality Theory in LP: The Furniture Factory Problem

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$$
\begin{aligned}
\max_{x_1,x_2} \quad & 45x_1 + 80x_2 \\
s.t. \quad & 5x_1 + 20x_2 \le 400 \\
& 10x_1 + 15x_2 \le 450 \\
& x_1, x_2 \ge 0
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$$

$$
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$$

- Properties:
  - **At optimality**, the optimal dual variables are the shadow prices of the resources

Duality Theory in LP: The Furniture Factory Problem

- The primal and dual formulations:

$$\begin{aligned} \max_{x_1,x_2} \quad & 45x_1 + 80x_2 \\ s.t. \quad & 5x_1 + 20x_2 \leq 400 \\ & 10x_1 + 15x_2 \leq 450 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$$\begin{aligned} \min_{\mu_1,\mu_2} \quad & 400\mu_1 + 450\mu_2 \\ s.t. \quad & 5\mu_1 + 10\mu_2 \geq 45 \\ & 20\mu_1 + 15\mu_2 \geq 80 \\ & \mu_1, \mu_2 \geq 0 \end{aligned}$$

- Properties:
  - **At optimality**, complementary slackness (CS) conditions hold

$$(400 - 5x_1^* - 20x_2^*) \times \mu_1^* = 0 \qquad (5\mu_1^* + 10\mu_2^* - 45) \times x_1^* = 0$$

$$(450 - 10x_1^* - 15x_2^*) \times \mu_2^* = 0 \qquad (20\mu_1^* + 15\mu_2^* - 80) \times x_2^* = 0$$

How to write the dual problem

Primal problem (or dual problem) Maximise	Dual problem (or primal problem) Minimise
Constraints:	**Variables:**
≤ form	≥ 0
= form	Unconstrained
≥ form	≤ 0
Variables:	**Constraints:**
≥ 0	≥ form
Unconstrained	= form
≤ 0	≤ form

How to write the dual problem

Primal problem (or dual problem) Maximise		Dual problem (or primal problem) Minimise
Constraints:		**Variables:**
≤ form	↔	≥ 0
= form	↔	Unconstrained
≥ form	↔	≤ 0
Variables:		**Constraints:**
≥ 0	↔	≥ form
Unconstrained	↔	= form
≤ 0	↔	≤ form

$$
\begin{aligned}
\max\ & 3x_1 + 2x_2 \\
s.t.\ & 2x_1 + x_2 \leq 100 \\
& x_1 + x_2 \geq 80 \\
& 4x_1 + 3x_2 = 240 \\
& x_1 \geq 0, x_2 \text{ free}
\end{aligned}
$$

$$\Downarrow$$

How to write the dual problem

Primal problem (or dual problem) Maximise		Dual problem (or primal problem) Minimise
Constraints:		**Variables:**
≤ form	⟷	≥ 0
= form	⟷	Unconstrained
≥ form	⟷	≤ 0
Variables:		**Constraints:**
≥ 0	⟷	≥ form
Unconstrained	⟷	= form
≤ 0	⟷	≤ form

$$\begin{aligned} \max\ & 3x_1 + 2x_2 \\ s.t.\ & 2x_1 + x_2 \le 100 \\ & x_1 + x_2 \ge 80 \\ & 4x_1 + 3x_2 = 240 \\ & x_1 \ge 0, x_2 \text{ free} \end{aligned}$$

⬇

$$\begin{aligned} \min\ & 100y_1 + 80y_2 + 240y_3 \\ s.t.\ & 2y_1 + y_2 + 4y_3 \ge 3 \\ & y_1 + y_2 + 3y_3 = 2 \\ & y_1 \ge 0 \\ & y_2 \le 0 \\ & y_3 \text{ free} \end{aligned}$$

$$
\begin{aligned}
\max\ & 3x_1 + 2x_2 \\
s.t.\ & 2x_1 + x_2 \le 100 \\
& x_1 + x_2 \ge 80 \\
& 4x_1 + 3x_2 = 240 \\
& x_1 \ge 0, x_2 \text{ free}
\end{aligned}
$$

$$
\begin{aligned}
\min\ & 100y_1 + 80y_2 + 240y_3 \\
s.t.\ & 2y_1 + y_2 + 4y_3 \ge 3 \\
& y_1 + y_2 + 3y_3 = 2 \\
& y_1 \ge 0 \\
& y_2 \le 0 \\
& y_3 \text{ free}
\end{aligned}
$$

Primal problem (or dual problem) Maximise		**Dual problem (or primal problem)** Minimise
Constraints:		**Variables:**
≤ form	↔	≥ 0
= form	↔	Unconstrained
≥ form	↔	≤ 0
Variables:		**Constraints:**
≥ 0	↔	≥ form
Unconstrained	↔	= form
≤ 0	↔	≤ form

How to write the dual problem

Primal problem (or dual problem)	Dual problem (or primal problem)
Maximise	Minimise
Constraints:	**Variables:**
≤ form	≥ 0
= form	Unconstrained
≥ form	≤ 0
Variables:	**Constraints:**
≥ 0	≥ form
Unconstrained	= form
≤ 0	≤ form

$$\min\ 4x_1 - 5x_2$$
$$s.t.\ 3x_1 + 2x_2 \geq 7$$
$$-x_1 + 6x_2 \leq 12$$
$$x_1 \geq 0,\ x_2 \leq 0$$

$$\Downarrow$$

How to write the dual problem

Primal problem (or dual problem)		Dual problem (or primal problem)
Maximise		Minimise
Constraints:		**Variables:**
≤ form	↔	≥ 0
= form	↔	Unconstrained
≥ form	↔	≤ 0
Variables:		**Constraints:**
≥ 0	↔	≥ form
Unconstrained	↔	= form
≤ 0	↔	≤ form

$$\begin{aligned} \min\ & 4x_1 - 5x_2 \\ s.t.\ & 3x_1 + 2x_2 \geq 7 \\ & -x_1 + 6x_2 \leq 12 \\ & x_1 \geq 0, x_2 \leq 0 \end{aligned}$$

$$\Downarrow$$

$$\begin{aligned} \max\ & 7y_1 + 12y_2 \\ s.t.\ & 3y_1 - y_2 \leq 4 \\ & 2y_1 + 6y_2 \geq -5 \\ & y_1 \geq 0 \\ & y_2 \leq 0 \end{aligned}$$

Duality Theory in LP

"Building on his results on matrix games and on his model of an expanding economy, von Neumann invented the theory of duality in linear programming when George Dantzig described his work in a few minutes, and an impatient von Neumann asked him to get to the point. Dantzig then listened dumbfounded while von Neumann provided an hourlong lecture on convex sets, fixed-point theory, and duality, conjecturing the equivalence between matrix games and linear programming."

----"Reminiscences about the origins of linear programming.". In Bachem, A.; Grötschel, M.; Korte, B. (eds.). *Mathematical Programming The State of the Art: Bonn 1982*. Berlin, New York: Springer-Verlag. pp. 78–86.

Setting up Microsoft Excel™

Step 0: Link the solver libraries

MS Excel → Tools → Add-Ins… → Solver Add In
or (in office 2013):
MS Excel → File → Options → Add-Ins → Solver Add-In

Step 1. Set up the Problem data

Step 2. Set up the solver data

Step 3. Solve

Step 4. Study results

Diet Problem

A patient must pay special attention to his intake of vitamin A: he must have at least 5000 IUs vitamin A every day. Due to his location, he can only choose from three foods: corn, milk, and wheat bread. To maintain healthy, he must have at least 2000 calories per day, but not more than 2250. The cost of each food and its nutrition and availability are as follows.

Food	Cost/Serving	Vitamin A	Calories	Availability
Corn (x)	$0.18	107	72	10
Milk (y)	$0.23	500	121	10
Wheat bread (z)	$0.05	0	65	10

He is tight on budge and cannot spend too much money on food. What is the minimum cost he can spend on food to satisfy his nutritional requirements?

THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY | SCHOOL OF ENGINEERING

Industrial Engineering & Decision Analytics

Diet Problem

Decision variables:

- x: servings of corn;
- y: servings of milk;
- z: servings of wheat bread

Objective function: to minimize the cost ➔

$$0.18x + 0.23y + 0.05z$$

Nutritional constraints:

Min Vitamin A:	$107x + 500y \geq 5000$
Min Calories:	$72x + 121y + 65z \geq 2000$
Max Calories:	$72x + 121y + 65z \leq 2250$

Step 1. Set up problem data

min $0.18x + 0.23y + 0.05z$	[objective]
s.t. $107x + 500y \geq 5000$	[Vitamin A]
$72x + 121y + 65z \geq 2000$	[Min Calories]
$72x + 121y + 65z \leq 2250$	[Max Calories]
$10 \geq x, y, z \geq 0$,	[non-negativity & availability]

$0.18x + 0.23y + 0.05z$

E5 =B5*B3+C5*C3+D5*D3

	A	B	C	D	E	F
1						
2	Variables:	x	y	z		
3		0	0	0		
4						
5	Objective	0.18	0.23	0.05	0	
6						
7	Constraints					
8	Vitamin A	107	500		0	5000
9	Calories	72	121	65	0	2000
10						2250
11	Availability	10	10	10		

Step 1. Set up problem data

min $0.18x + 0.23y + 0.05z$	[objective]
s.t. $107x + 500y \geq 5000$	[Vitamin A]
$72x + 121y + 65z \geq 2000$	[Min Calories]
$72x + 121y + 65z \leq 2250$	[Max Calories]
$10 \geq x, y, z \geq 0,$	[non-negativity & availability]

$107x + 500y$

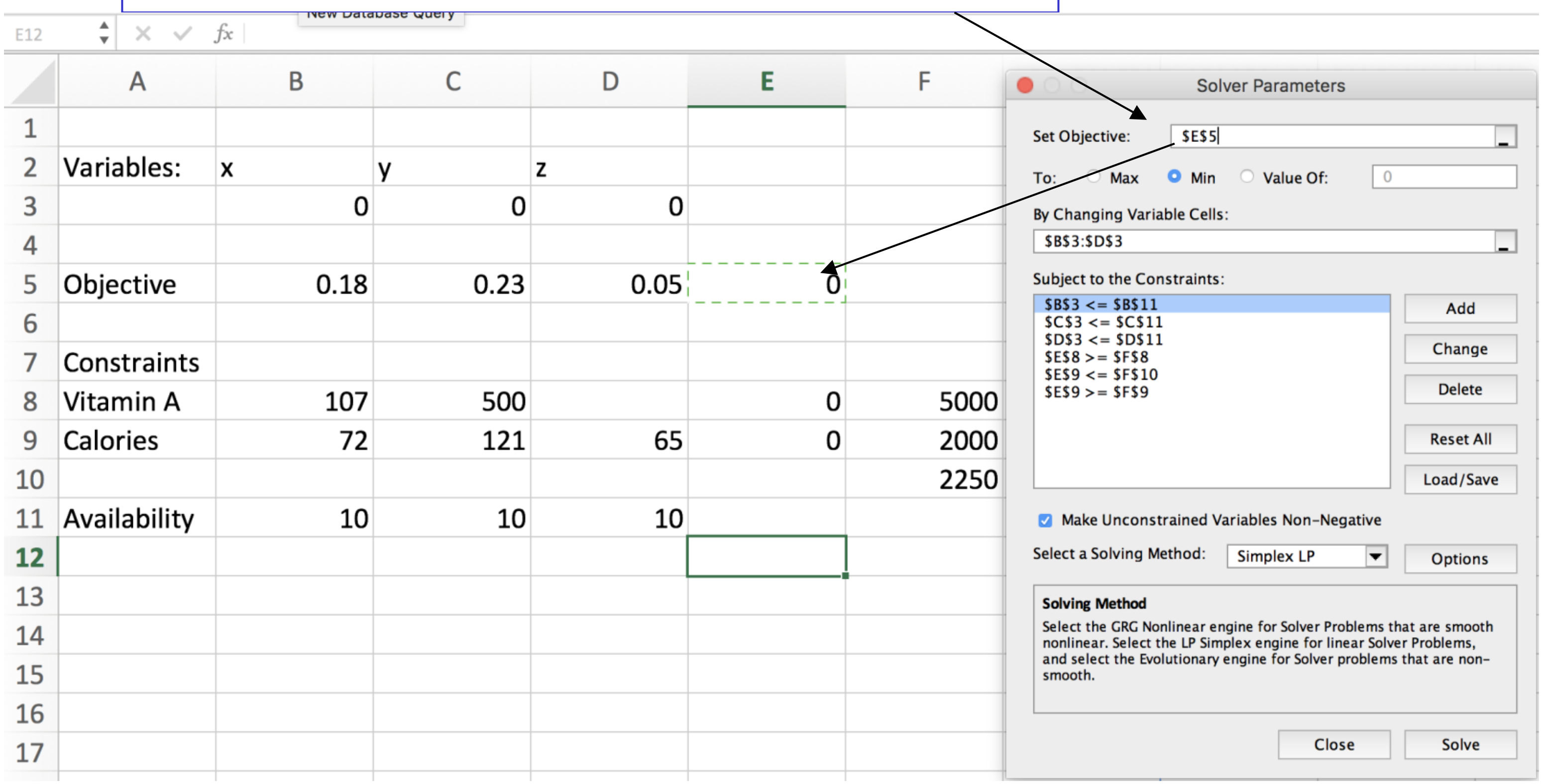
E8 =B8*B3+C8*C3

	A	B	C	D	E	F
1						
2	Variables:	x	y	z		
3		0	0	0		
4						
5	Objective	0.18	0.23	0.05	0	
6						
7	Constraints					
8	Vitamin A	107	500		0	5000
9	Calories	72	121	65	0	2000
10						2250
11	Availability	10	10	10		

Step 2. Set up the Solver Data

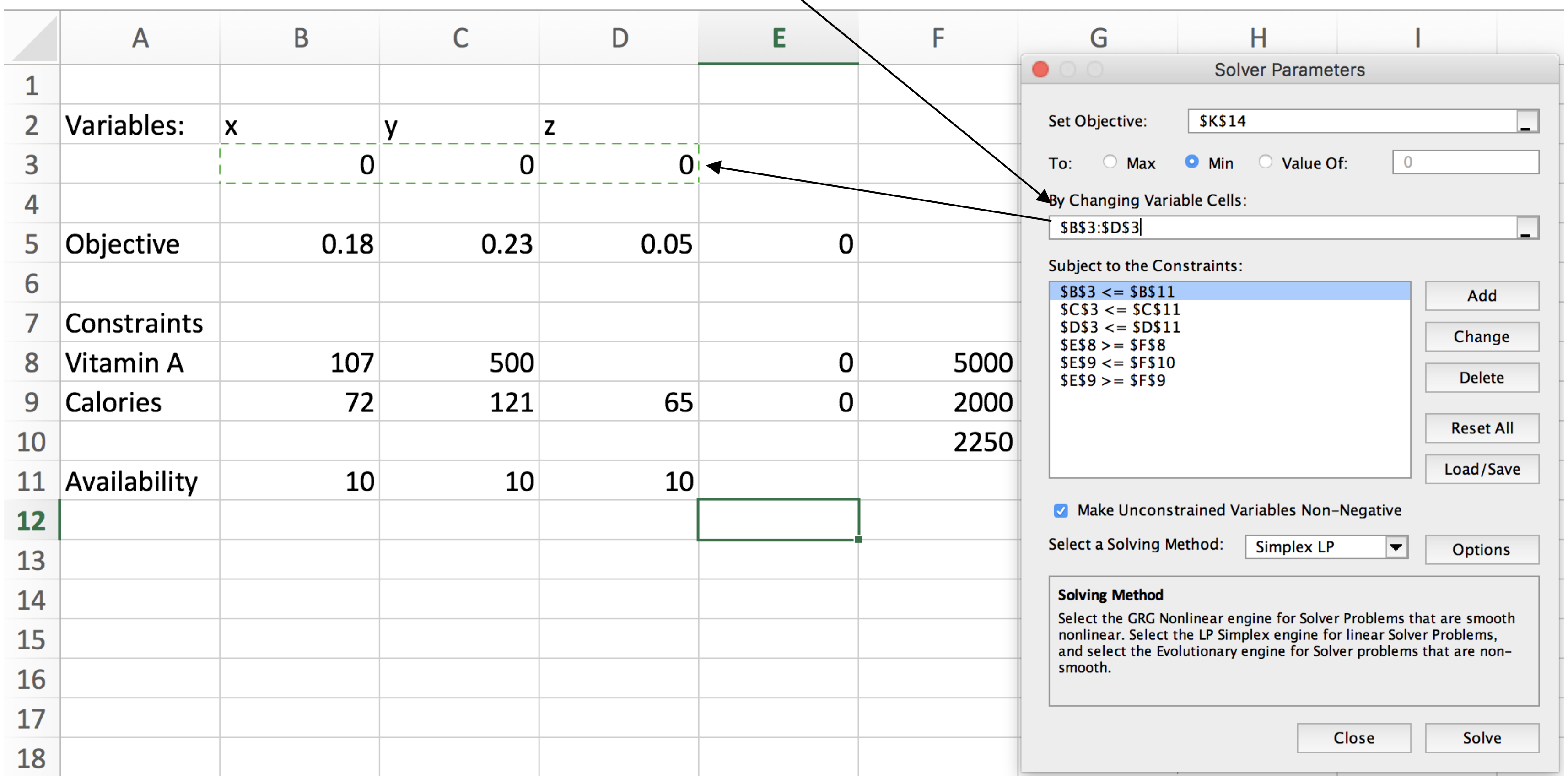
Step 2.1 Tools → Solver…

Step 2.2 Objective function



Step 2. Set up the Solver Data

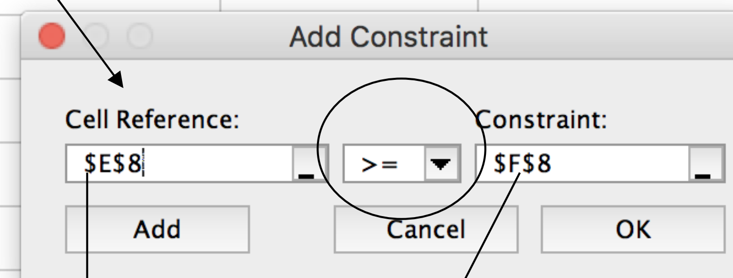
Step 2.3 Specifying the variables



Step 2. Set up the Solver Data

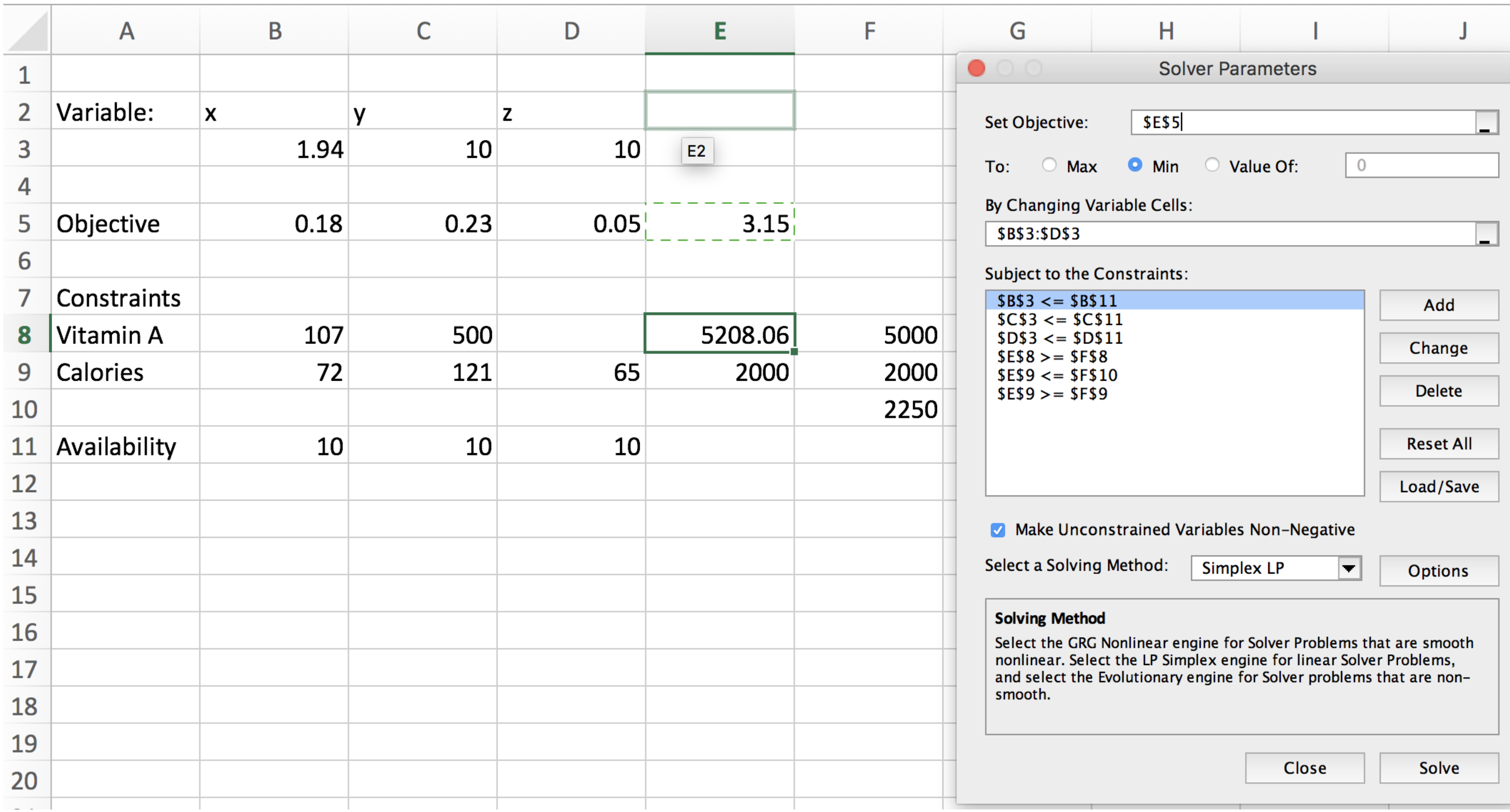
Step 2.4 Specifying the constraints

	A	B	C	D	E	F	G
1							
2	Variables:	x	y	z			
3		0	0	0			
4							
5	Objective	0.18	0.23	0.05			
6							
7	Constraints						
8	Vitamin A	107	500		0	5000	
9	Calories	72	121	65	0	2000	
10						2250	
11	Availability	10	10	10			



Step 3. Solve

Complete problem specification



Step 4. Study the results

How much corn

How much milk

How much bread

Min cost

2	Variable:	x	y	z		
3		1.94	10	10		
4						
5	Objective	0.18	0.23	0.05	3.15	
6						
7	Constraints					
8	Vitamin A	107	500		5208.06	5000
9	Calories	72	121	65	2000	2000
10						2250
11	Availability	10	10	10		

How much Vitamin A

How much calories

Practice

Failsafe Electronics Corporation primarily manufactures four highly technical products, which it supplies to aerospace firms that hold NASA contracts. Each of the products must pass through the following departments before they are shipped: wiring, drilling, assembly, and inspection. The time requirements in each department (in hours) for each unit produced and its corresponding profit value are summarized in this table. What is the optimal Product mix?

Product	Wiring	Drilling	Assembly	Inspection	Unit Profit
XJ201	.5	3	2	.5	$9
XM897	1.5	1	4	1.0	$12
TR29	1.5	2	1	.5	$15
BR788	1.0	3	2	.5	$11
Available hours	1500	2350	2600	1200	